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Title

**Detection And Isolation Of Antimicrobial Producer From Soil
Collected From Different Areas In Sudan - 2025**

A thesis Submitted in Partial Fulfillment for requirement of B.Sc. in Medical
laboratory sciences (Microbiology and Immunology)

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آية قرآنية

اللَّهُ لَا إِلَهَ إِلَّا هُوَ الْحَيُّ الْقَيُّومُ
لَا تَأْخُذُهُ سِنَّةٌ وَلَا نَوْمٌ لَهُ مَا فِي السَّمَوَاتِ وَمَا فِي الْأَرْضِ
مَنْ ذَا الَّذِي يَشْفَعُ عِنْدَهُ إِلَّا بِإِذْنِهِ يَعْلَمُ مَا بَيْنَ أَيْدِيهِمْ
وَمَا خَلْفَهُمْ وَلَا يُحِيطُونَ بِشَيْءٍ مِّنْ عِلْمِهِ إِلَّا بِمَا شَاءَ
وَسِعَ كُرْسِيُّهُ السَّمَوَاتِ وَالْأَرْضَ وَلَا يَئُودُهُ حِفْظُهُمَا
وَهُوَ الْعَلِيُّ الْعَظِيمُ ﴿٢٥٥﴾

Dedication

*It is with our genuine gratitude that we dedicate this study
to our fathers...*

to our mothers...

to our dear sisters and brothers...

*and to all our teachers and colleagues...and to all passionate researchers whose
commitment to science continuous to inspire.*

Acknowledgement

Firstly, we are deeply grateful to **Allah**, who enabled us to complete this work. We would like to thank all those whose assistance proved to be a milestone in the accomplishment of our goal.

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She has always been eager to guide us, listen to our suggestions and queries throughout the study, and ensured that we gained the maximum benefit from the program.

Abstract

Background : Antibiotic resistance is a growing global crisis, leading to renewed efforts in discovering new antibiotics, particularly from natural sources. Historically, soil has been a primary source of clinically significant antibiotics like streptomycin and penicillin. Microorganisms in soil, especially *Bacillus*, *Streptomyces*, and fungi, produce antibiotics to survive in competitive environments. Due to the urgent need for new antimicrobial agents, soil continues to be explored for novel antibiotic-producing organisms.

Aim: To detect antibiotic-producing microorganisms from soil collected from different regions and see the effect in the clinical isolated bacteria spp.

Method: Soil samples were cultured on different media (nutrient agar, yeast extract agar, glucose-enriched agar). Streaking and overlay techniques with clinical isolates to observe inhibition zones. Dutch Streak Method and Mueller-Hinton agar were used to confirm antimicrobial activity. Identification involved gram staining, KOH prep, litmus milk test, bile aesculin test, and motility test.

Results: Microorganisms with antibiotic activity were found in samples from Nuri, Alabar, Atbara, Port Sudan, and Shandi. Antimicrobial activity was observed against clinical strains: *S. aureus* (up to 52 mm inhibition), *P. aeruginosa*, *K. pneumoniae*, *E. coli*, and *Bacillus* spp. Identified producers included fungi, *Bacillus* spp., and suspected *Clostridium* and *Listeria*.

Statistical analysis showed: Significant differences in inhibition zones between soil isolates and controls (amoxicillin/doxycycline) for most pathogens. Some soil isolates, like Nuri 1 large, demonstrated strong activity comparable to commercial antibiotics.

Conclusions: Soil from various regions harbors promising antibiotic-producing microorganisms. *Bacillus* spp. and certain fungi showed antimicrobial effects against key clinical pathogens. Though generally less effective than commercial antibiotics, some isolates had comparable inhibition zones. Soil remains a viable reservoir for discovering new antibiotics, crucial in the fight against resistant pathogens.

المستخلص

الخلفية: مقاومة المضادات الحيوية هي أزمة عالمية متزايدة، مما يؤدي إلى جهود متجددة في اكتشاف مضادات حيوية جديدة، وخاصة من مصادر طبيعية. تاريخياً، كانت التربة المصدر الرئيسي لمضادات حيوية ذات أهمية سريرية مثل الستربتوميسين والبنسلين. تنتج الكائنات الدقيقة في التربة، وخاصة الباسيلاس والستربتوميسيس والفطريات، مضادات حيوية للبقاء في بيئات تنافسية. بسبب الحاجة الملحة لوكلاء مضادة للميكروبات جديدة، تستمر التربة في استكشاف الكائنات المنتجة لمضادات حيوية جديدة.

الهدف: اكتشاف الكائنات الدقيقة المنتجة للمضادات الحيوية من التربة التي تم جمعها من مناطق مختلفة ومراقبة التأثير على أنواع البكتيريا المعزولة سريرياً.

الطريقة: تم زراعة عينات التربة على وسائط مختلفة (أغار المغذيات، أغار مستخلص الخميرة، أغار غني بالجلوكوز). تم استخدام تقنيات التحريك والتغطية مع العزلات السريرية لملاحظة مناطق التثبيط. تم استخدام طريقة التحريك الهولندية وأغار مولر-هينتون لتأكيد النشاط المضاد للميكروبات. شمل التعريف صبغ جرام، وتحضير KOH، واختبار الحليب بمؤشر الغضب، واختبار البيل أيسولين، واختبار الحركة.

النتائج: وُجدت كائنات دقيقة ذات نشاط مضاد للمضادات الحيوية في عينات من النوري، الأبار، عطبرة، بورتسودان، وشندي. تم ملاحظة نشاط مضاد للميكروبات ضد السلالات السريرية: *S. aureus* (حتى 52 مم تثبيط)، *P. aeruginosa*، *K. pneumoniae*، *E. coli*، و *Bacillus spp*. وشملت المنتجات المحددة الفطريات و *Bacillus spp*، و *Clostridium* و *Listeria* المشتبه بهم.

أظهرت التحليلات الإحصائية: اختلافات ملحوظة في مناطق التثبيط بين عزلات التربة والعينات الضابطة (أموكسيسيلين / دوكسي سيكلين) لمعظم مسببات الأمراض. أظهرت بعض عزلات التربة، مثل نوري 1 كبيرة، نشاطاً قوياً قابلاً للمقارنة مع المضادات الحيوية التجارية.

الاستنتاجات: يحتوي التربة من مناطق مختلفة على كائنات دقيقة منتجة للمضادات الحيوية واعدة. أظهرت أنواع البكتيريا *Bacillus* وبعض الفطريات تأثيرات مضادة للميكروبات ضد مسببات الأمراض السريرية الرئيسية. على الرغم من أنها عموماً أقل فعالية من المضادات الحيوية التجارية، إلا أن بعض العزلات كانت لديها مناطق تثبيط مقارنة. تظل التربة مصدراً قابلاً للاستخدام لاكتشاف مضادات حيوية جديدة، وهو أمر حاسم في مواجهة مسببات الأمراض المقاومة.

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List Of Abbreviations

Abbreviation	Full Form
AMR	Antimicrobial Resistance
CFU	Colony Forming Unit
DW	Distilled Water
G+ve	Gram-positive
G-ve	Gram-negative
KOH	Potassium Hydroxide
MH	Mueller-Hinton
NA	Nutrient Agar
SDA	Sabouraud Dextrose Agar
YE	Yeast Extract
DNA	Deoxyribonucleic Acid
RNA	Ribonucleic Acid
spp.	Species (plural)
MIC	Minimum Inhibitory Concentration
µg	Microgram
mm	Millimeter

Chapter One

Chapter One:

1. Introduction and Literature Review

1.1 Introduction

Antibiotics are chemical substances produced naturally by microorganisms, or synthetically derived, that inhibit the growth of or destroy other microorganisms, particularly bacteria. They are primarily used to treat bacterial infections by either killing the bacteria (bactericidal) or inhibiting their growth and reproduction (bacteriostatic). Most clinically important antibiotics have been derived from soil microorganisms such as *Streptomyces*, *Bacillus*, and certain fungi like *Penicillium*. The history of antibiotic production began in the early 20th century, revolutionizing medicine. Although natural remedies were used for centuries, the true scientific breakthrough came in 1928, when Alexander Fleming discovered penicillin from the fungus *Penicillium notatum*. However, it wasn't until the 1940s that penicillin was mass-produced and used widely during World War II, saving countless lives. Following this, a “golden age of antibiotics” emerged between the 1940s and 1960s, during which many antibiotics were discovered—mostly from soil-dwelling microorganisms like *Streptomyces*. Examples include streptomycin, tetracycline, erythromycin, and chloramphenicol. Over time, semi-synthetic and synthetic antibiotics were developed to improve activity and overcome resistance (Madigan et al., 2018) .

However, the rise of antibiotic resistance in recent decades has renewed interest in discovering new antibiotics, especially from natural sources like soil and marine environments. Soil is a rich and diverse ecosystem that plays a crucial role in microbiology, especially in the discovery of antibiotics. It contains a vast array of microorganisms, including bacteria, fungi, and actinomycetes, which interact and compete for nutrients and space. This competitive environment drives many microbes to produce secondary metabolites—such as antibiotics—to inhibit or kill their rivals. (Bérdy, 2012)

Historically, soil has been the primary source for most clinically used antibiotics. For example, *Streptomyces* species, commonly found in soil, are responsible for producing over two-thirds of all natural antibiotics, including streptomycin, tetracycline, and erythromycin. The study of soil microbiology continues to be essential for identifying novel antibiotic compounds to combat rising antimicrobial resistance. The rapid emergence of antibiotic-resistant pathogens poses a significant threat to global public health, creating an urgent need for the discovery of novel antimicrobial agents. (Genilloud, 2017)

Soil, a complex and dynamic ecosystem, harbors a vast diversity of microorganisms, many of which are known producers of bioactive secondary metabolites, including antibiotics. Historically, soil-dwelling microbes, particularly species of *Streptomyces*, *Bacillus*, and *Penicillium*, have been pivotal in the discovery of clinically important antibiotics such as streptomycin, tetracycline, and penicillin. Microbial competition in the soil environment often leads to the production of antimicrobial compounds as a survival strategy, making soil a rich reservoir for antibiotic-producing organisms. (Bérdy, 2012)

In recent decades, the world has witnessed a growing health crisis—antibiotic resistance. Once hailed as miracle drugs, antibiotics are now losing their effectiveness as bacteria evolve and develop resistance to them. This alarming trend threatens to undo decades of medical progress, making common infections harder—and sometimes impossible—to treat. The misuse and overuse of antibiotics in humans, animals, and agriculture have accelerated the spread of resistant strains, turning once-treatable diseases into deadly threats. As antibiotic resistance continues to rise globally, it poses a serious challenge to public health, food security, and the future of modern medicine. The global health community is facing one of the most serious threats of the 21st century: antibiotic resistance. This phenomenon occurs when bacteria evolve mechanisms to withstand the drugs designed to kill them. What was once considered a medical miracle—antibiotics—has now become a double-edged sword. The overuse and misuse of antibiotics in healthcare, veterinary medicine, and agriculture have led to the rapid emergence of resistant bacteria, also known as "superbugs." These resistant strains render standard treatments ineffective, leading to longer hospital stays, higher medical costs, and increased mortality (Ventola, 2015)

1.2 Literature Review

1.2.1 Soil as a Reservoir of Antimicrobial-Producing Microorganisms

1.2.1.1 Introduction to antibiotics : Historically, the majority of clinically used antibiotics have originated from soil microorganisms, particularly from the genera *Streptomyces*, *Bacillus*, and *Penicillium*. These microbes naturally produce antibiotics as secondary metabolites to compete with other organisms in their environment (Kumar et al., 2014). Actinomycetes, especially *Streptomyces*, are responsible for producing over two-thirds of all known antibiotics, making soil a critical reservoir in the fight against infectious diseases (Sharma et al., 2021).

However, the emergence of antimicrobial resistance (AMR) has created an urgent global demand for new antibiotics. Pathogens such as *Escherichia coli*, *Klebsiella pneumoniae*, *Staphylococcus aureus*, and *Pseudomonas aeruginosa* are increasingly resistant to commonly used drugs. Soil microbiomes are once again in the spotlight as promising sources of novel antimicrobial agents. For instance, (Sandhya et al. , 2012) successfully isolated *Bacillus* strains from local soils that showed strong antimicrobial activity against several resistant bacteria. Similarly, (Khalid et al. , 2016) reported over 50 soil isolates, many of which demonstrated broad-spectrum inhibition, highlighting soil as a productive environment for new antibiotics.

1.2.1.2 Microbial diversity in soil (bacteria, fungi, actinomycete) : All reviewed studies confirmed that soil harbors a wide range of microorganisms including *Bacillus*, *Streptomyces*, Actinomycetes, *Kocuria*, and fungi such as *Rhizopus stolonifer* (Microorganisms, 2013)

1.2.1.3 Ecological factors influencing antimicrobial activity in soil microbes: emphasized that soils with minimal human interference (e.g., wasteland alkaline soils) yielded more potent antimicrobial producers compared to agricultural soils, suggesting ecological stress enhances secondary metabolite production. (Sharma et al. , 2021)

1.2.1.4 Importance of soil types and locations (e.g., rhizosphere, forest soil, etc) : noted that agricultural soils across various regions (India and Egypt) supported actinomycete

diversity (Kumar et al. , 2014) and (Elbendary et al. , 2018). Wasteland soils and compost-rich garden soils were found to be especially promising (Sharma et al., 2021).

1.2.2 Types of antimicrobials isolated from the soil

1.2.2.1 Aminoglycosides (e.g., Streptomycin, Gentamicin) : Traditional soil-derived antibiotics such as streptomycin originate from *Streptomyces griseus*, while gentamicin comes from *Micromonospora* spp. Modern research frequently isolates *Streptomyces* strains with broad activity—suggesting this class remains abundant in soil . (e.g., SCA7 from Kumar et al., 2014)

1.2.2.2 Tetracyclines : These ribosomal inhibitors (e.g., tetracycline, oxytetracycline) are produced by *Streptomyces rimosus* and *S. aureofaciens*. Reported soil-derived *Bacillus* strains resistant to tetracycline, implying environmental exposure to tetracycline-like molecules or resistance gene pools (Sandhya et al. , 2012).

1.2.2.3 Chloramphenicol : Synthesized by *Streptomyces venezuelae*, this classic broad-spectrum antibiotic is frequently associated with soil actinomycetes. Strong activity seen in wasteland isolates hints at its potential presence (Sharma et al., 2021; inhibition >20 mm) .

1.2.2.4 Rifamycins : Derived from *Streptomyces rifamycinica*, rifamycins target RNA polymerase. While not specifically identified in the provided studies, actinobacterial isolates suggest the presence of this antibiotic type in soil . (Elbendary et al., 2018; Khalid et al., 2016)

1.2.3 Common Genera of Antimicrobial Producers Isolated from Soil

1.2.3.1 Actinomycetes : Identified as dominant antimicrobial producers in multiple studies, particularly in (Kumar et al., 2014) and (Sharma et al., 2021), with *Streptomyces* sp. SCA 7 showing strong bioactivity.

1.2.3.2 Bacillus spp. : Found frequently in all studies. (Khalid et al., 2016) isolated 39 *Bacillus* strains, including *B. subtilis* and *B. licheniformis*. (Sandhya et al., 2012) confirmed *Bacillus* spp. produced large inhibition zones.

1.2.3.3 Pseudomonas spp. : Though commonly isolated (Microorganisms, 2013), they rarely exhibited antimicrobial activity. Most isolates were non-producers in this genus.

1.2.4 Methods Used for Detection and Isolation

1.2.4.1 Culture-based methods (e.g., cross-streaking, agar well diffusion) : All studies used culture-based screening. Agar well diffusion was common (Sandhya et al., 2012; Elbendary et al., 2018). Cross-streaking was noted in (Microorganisms , 2013)

1.2.4.2 Molecular and biochemical identification techniques: Biochemical tests were used widely (Sandhya et al., Khalid et al.), while (Kumar et al., 2014) used 16S rDNA sequencing. (Elbendary et al., 2018) used the VITEK 2 system for high-resolution ID.

1.2.5 Antimicrobial Compounds and Their Targets

1.2.5.1 Mechanisms of action: Not directly specified in the reviewed studies, but zones of inhibition suggest inhibition of cell wall or protein synthesis. (Kumar et al., 2014) mentioned possible bioactive phenolic compounds.

1.2.5.2 Spectrum of activity (Gram-positive, Gram-negative, fungi) : Most isolates were active against both Gram-positive (*S. aureus*, *B. subtilis*) and Gram-negative (*E. coli*, *P. aeruginosa*) bacteria. Some, like in (Elbendary et al., 2018) showed antifungal activity against *C. albicans* and *A. niger*.

1.3 Rationale

Soil is a highly diverse and competitive environment that harbors numerous microorganisms, making it an excellent source for discovering antibiotic-producing species. Microorganisms in soil, particularly *Streptomyces*, *Bacillus*, and *Penicillium* species, produce antibiotics as a survival strategy to outcompete others for nutrients and space.

With the rise of antibiotic resistance, there is an urgent need to explore new sources like soil to find alternative or more effective antimicrobial agents.

1.4 Objective

1.4.1 General Objective: detection of antibiotic producer from soil collected from different areas

1.4.2 Specific objectives:

- To detect the effect of antibiotic produced from soil on clinical isolated Gram positive bacterial spp. (*Staphylococcus aureus* , *Bacillus spp.*)
- To detect the effect of antibiotic produced from soil on clinical isolated Gram negative bacterial spp (*Klebsiella spp.* , *Escherichia coli*)
- To detect the effect of antibiotic produced from soil on clinical isolated *Pseudomonas aeruginosa*

Chapter Two

Chapter Two:

2. Materials And Methods

2.1 Study design: Cross-sectional study

2.2 Study area: This study will be performed in Mashriq University lab and the sample will be collected from different areas in Sudan.

2.3 Study period: April /2025 – July /2025

2.4 Study population: Soil.

2.5 Lab location : Mashriq University.

2.6 Inclusion criteria: Soil from clean area.

2.7 Exclusion criteria: Soil from dirty area.

2.8 Materials and Equipments : Cork borer , Automated pipette , incubator , hot air oven , autoclave , sensitive balance and wire loops.

2.9 Sample size : A total of 8 samples , was collected from different areas which are Nuri, Al-Duwaym, Shendi, Atbara, Port Sudan 1, Port Sudan 2, Sikka Sitta, and Al-Abar .

2.10 Sample Type and collection: soil is collected in sterile bags under good handling from clean area.

2.11 Data collection Tool: From previous studies.

2.12 Quality control: Group of antibiotics (Approval antibiotics)

2.13 Ethical consideration: The sample is collected from soil.

2.14 Methodology

2.14.1 Soil Sample Collection:

Soil samples were collected from eight different locations across Sudan to ensure geographic and environmental diversity. The locations included: Nuri, Al-Duwaym, Shendi, Atbara, Port Sudan 1, Port Sudan 2, Sikka Sitta, and Al-Abar.

2.14.2 Initial Dilution and Culturing on Nutrient Agar:

Two dilutions were prepared from each sample:

- 1gram of soil in 10 ml of distilled water
- 2grams of soil in 10 ml of distilled water

These were inoculated onto Nutrient Agar (NA) plates as three types for each soil sample (1g / 2g / mix) and incubated at 37c for two weeks. Then we seeded staphylococcus aureus from nutrient broth in each plate .However, no bacterial growth was observed, suggesting the medium or conditions were not suitable for the organisms present.

2.14.3 Culturing on Yeast Extract Agar:

To enhance the chances of fungul recovery, Yeast Extract Agar was prepared for each soil specimen that reprepared in the formula:

- 5grams of soil in 10 ml of distilled water
- 10grams of soil in 10 ml of distilled water

Then we inoculated it in the same mannar (5g / 10g / Mix) and incubated at 37c for one week. After one week of incubation, a few colonies appeared so we incubated it for another week to allow if there are slow grower microorganisms to grow. After one week , we seeded Staphylococcus aureus into melted nutrient agar and poured in each plate of the soil specimen and incubated for one day, then we noted there is no clear growth on the media.

2.14.4 Culturing on Glucose-Enriched Nutrient Agar:

To promote better microbial growth, Nutrient Agar was supplemented with glucose in the formula : 3g of glucose dissolved into 270 ml of melted nutrient agar

The specimens were reprepared in higher concentration of soil (10g of soil in 10 ml of DW for each specimen), followed by further 10 folds dilution for each specimen (1ml of each soil suspension into 9ml of DW). These modified samples were inoculated into the glucose-enriched agar, incubated for 24hrs at 37⁰C aerobically resulting in a notable increase in microbial colony growth. A broth culture of Staphylococcus aureus was seeded onto multed Nutrient Agar and poured on each cultured media (covered the colonies of each specimen).

After two days of incubation, several soil-derived colonies showed clear zones of inhibition, indicating antimicrobial activity, From each colony showed inhibition zone we stained it with gram stain and confirmed the fungul organisms by using 10% KOH seen in **Table 2.1**

NO	Soil specimen	Gram stain	10 %KOH
1	Portsudan 1 (large)	No seen	Sporangiophore
2	Portsudan 1 (small)	Gram positive bacili in chain	No seen
3	Nuri 1 (large)	Gram positive bacili / coccobacilli	No seen
4	Nuri 1 (small)	No seen	Aseptate hyphae / conidiophore
5	Nuri 2	Gram positive bacili in single (subterminal spore seen)	No seen
6	Al-abar	Gram positive bacili	Aseptate hyphae / mycelium

Table 2.1 : Illustrates the gram reaction and KOH examination for each colony suspected with antimicrobial effect from the glucose enriched nutrient agar

2.14.5 Purification step:

After another culture of diluted soil specimens and poured with melted nutrient agar seeded with *Staphylococcus aureus* , we collected new colonies with antimicrobial activity we subcultured from each colony was purified in Sabroud dextrose agar and incubated for 24hrs at 37°C , aerobically . After the incubation , we noted the colonial morphology for each colony , stained with gram stain and detected the presence of fungul by 10% fresh prepared KOH and then stored for the further tests. see results in **Table 2.2**

NO	Soil specimen	Colonial morphology	Gram stain	10% KOH
1	Nuri 1	Large , round , mucoid , irregular , high convex.	Gram positive bacilli / coccobacili	Aseptate hyphyae , conidiophore
2	Nuri 2	Small , round , regular , low convex.	Gram positive bacilli with terminal spore .	No seen
3	Portsudan 1	White , small , irregular , low convex.	Gram positive bacilli .	Sporangiophore
4	Alabar	White , round regular smooth , small , low convex.	Gram positive bacili	Aseptate hyphae
5	Shandi	White , small , irregular , low convex.	Gram positive bacili	No seen

Table 2.2 : Illustrates the colonial morphology , gram reaction and KOH examination for each colony purified from another confirmatory susceptibility test

2.14.6 Confirmation of antimicrobial activity via Dutch Streak Method:

To confirm the inhibition zones had seen due antimicrobial production, the Dutch streak method was used. The noted purified colonies was streaked horizontally on a Yeast extract agar plate. Four test bacteria were streaked vertically:

- Staphylococcus aureus
- Bacillus spp.

- *Pseudomonas aeruginosa*
- *Escherichia coli*

After incubation, distinct zones of inhibition were observed . The experiment was repeated three times to ensure reproducibility, and consistent results confirmed the isolate's antibiotic activity. Seen in **Fig 4.6.1**

2.14.7 Confirmatory Test for Antibiotic-Producing Microorganisms

After confirming the presence of antibiotic-producing microorganisms in selected soil samples, a confirmatory sensitivity assay was conducted using Mueller-Hinton agar supplemented with glucose (in the formula 3g of glucose dissolved in 270ml of melted Mueller hinton agar) . A total of 20 agar plates were prepared.

For the first ten plates, wells were made using a cork borer (see in the **Appendix**) with a diameter of 10 mm. Five different pathogenic bacterial strains were used as test organisms: *Staphylococcus aureus*, *Bacillus spp.* , *Klebsiella spp.* , *Escherichia coli*, and *Pseudomonas aeruginosa*. The bacterial suspensions were standardized using the McFarland standard, diluted accordingly, and each strain was swabbed onto two separate plates, resulting in two plates per bacterial species.

In each of these ten plates, seven wells were made and labeled with numbers 1 to 10 with A (Amoxicillin 5ug) and D (Doxycycline 10ug) - in the same media to compare them in the same conditions - which is prepared , corresponding to different isolates of antibiotic-producing microorganisms (Sikka sitta [2] , Atbara , Shandi , Alabar [2] , Portsudan1 , Nuri[3]) and we added 100ul from each antimicrobial producer suspension to the holes that numbered according for each soil specimen . Thus, each bacterial species was tested against ten different isolates. The cultured plates were incubated at 37°C aerobically in upside down incubation method . After one day of incubation we noted the inhibition zone that measured by using ruler as the below **Table 2.3** :

NO	Bacteria spp /antimicrobial producer site and control antibiotic	G +ve bacillus spp	Klebsiella pneumoniae	Escherichia coli	Pseudomonas aeruginosa	Staphylococcus aureus
1	Sikka sita	18 mm	0 mm	0 mm	0 mm	0 mm
2	Atbara	20 mm	20 mm	18 mm	18 mm	22 mm
3	Shandi	18 mm	16 mm	16 mm	22 mm	16 mm
4	Alabar 1	24 mm	26 mm	14 mm	16 mm	20 mm
5	Alabar 2	24 mm	20 mm	18 mm	16 mm	20 mm
6	Portsudan 1	16 mm	0 mm	0 mm	20 mm	18 mm
7	Nuri 1 small	28 mm	18 mm	16 mm	22 mm	18 mm
8	Nuri 1 large	28 mm	18 mm	20 mm	20 mm	52 mm
9	Amoxicillin 5 ug	38 mm	48 mm	0 mm	22 mm	40 mm
10	Doxycycline 10 ug	40 mm	24 mm	40 mm	42 mm	52 mm

Table 2.3 : Illustrates the measured inhibition zones due the antimicrobial effect on the clinical isolated bacterial spp.

Note : we excluded two specimens from the cultured (sikka sitta1 / nuri2) due no antimicrobial activity seen from them .

- ✓ For the remaining ten plates, the cross-streak method (**modified "Dutch method"**) was employed. A horizontal streak of each test bacterium was made across the center of the plate, and vertical streaks of the isolated antibiotic-producing microorganisms were then applied perpendicular to the test strain streaks. This process was repeated across the remaining nine plates , then incubated at 37°C aerobically for 24 hrs.

In the next day the noted results :

1. Antimicrobial producer from Alabar 2 showed antimicrobial activity on *Bacillus* spp.
2. Antimicrobial producer from Atbara showed antimicrobial activity on *Pseudomonas* spp. and *Bacillus* spp.
3. Antimicrobial producer from Nuri 1 small showed antimicrobial activity on *E. coli* and *pseudomonas* spp.
4. Antimicrobial producer from Alabar 1 showed antimicrobial activity on *Bacillus* spp. , *Klebsiella* spp. and *Pseudomonas* spp.
5. Antimicrobial producer from Shandi showed antimicrobial activity on *Bacillus* spp. , *Klebsiella* spp. , *E. coli* and *Staph. aureus*

Clear zones of inhibition were observed in both methods, indicating effective antimicrobial activity. The results were consistent and demonstrated that several of the isolated microorganisms exhibited promising antibiotic potential. Seen in fig (2)

2.14.8 Identification of Antimicrobial Producer:

For identification of antimicrobial producer the tests used are illustrated in **Table 2.4** :

NO	Antimicrobial producer site	Gram stain , colonial morphology	Litmus milk reduction test	Bile aesculin test	Motility test
1	Sikka sita	Fungi [white , large , round , cottony , hairy colony]			
2	Atbara	G+ bacilli , [white , small , smooth , round colony]	Negative	Negative	Motile

3	Shandi	G+ bacilli and fungi , [white , small , regular colony / other cottony colony]	Positive	Negative	Motile
4	Alabar 1	G+ bacilli with terminal spore [pale yellow , small , regular , smooth colony]	Stormy clot	Negative	Motile
5	Alabar 2	G+ bacilli [white , large , smooth , thin margin , irregular colony]	Negative	Negative	Motile
6	Portsudan 1	G+ bacilli [white , round , small colony]	Negative	Negative	Motile
7	Nuri 1 small	G+ bacilli [white , small , mucoid colony]	Stormy clot	Negative	Motile
8	Nuri 1 large	G+ bacilli [white , large , round , mucoid]	Stormy clot	Negative	Motile

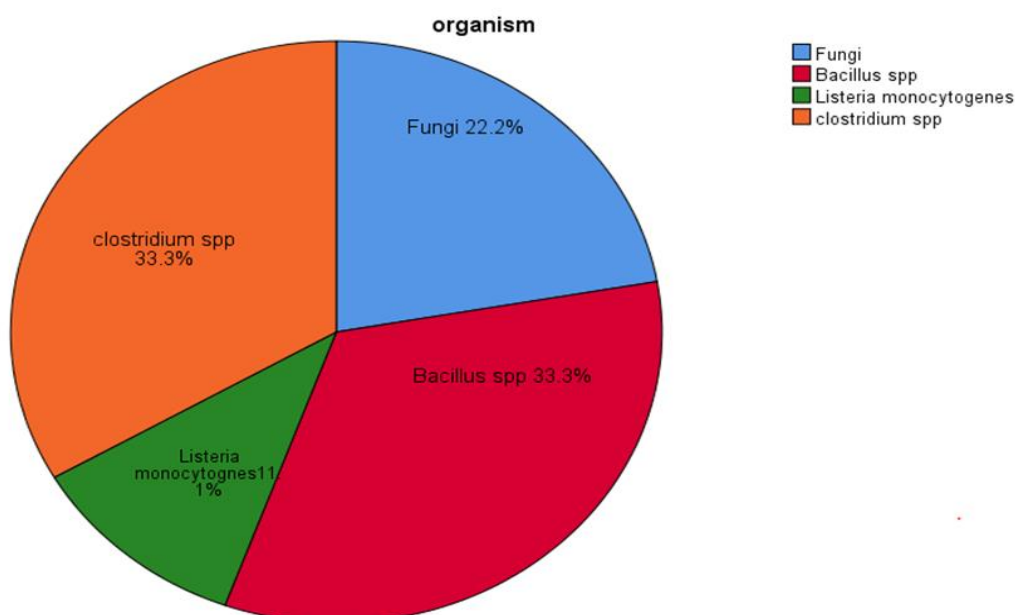
Table 2.4 : Illustrate the results of the biochemical tests (**Litmus Milk Reduction , Bile Esculine Hydrolysis and Motility**) used for identification of antimicrobial producer from soil specimens

Result:

According to the **colonial morphology** , **gram stain** and **10% KOH** and **biochemical tests** . we noted that the antimicrobial producers from soil commonly as seen in **Table 2.5**

NO	Soil location	Fungi	Bacillus spp.	Listeria monocytogenes	Clostridium spp.
1	Sikka sita	1	-	-	-
2	Atbara	-	1	-	-
3	Shandi	-	-	1	-
4	Alabar 1	1	-	-	1
5	Alabar 2	-	1	-	-
6	Portsudan1	-	1	-	-
7	Nuri 1 S	-	-	-	1
8	Nuri 1 L	-	-	-	1

Table 2.5 : Illustrates the common antimicrobial producers from soil due to the colonial morphology , gram stain and 10% KOH and biochemical tests. As showed in the below **pie chart** :



Chapter three

Chapter Three:

3. Results :

This study was conducted to investigate the presence and effectiveness of antibiotic-producing microorganisms isolated eight soil samples collected across various locations in Sudan. The Atbara isolate inhibited all five tested pathogens with inhibition zones ranging from 18 mm(40%) to 22 mm(20%) [**Chart 3.1**]. The Alabar 1 and 2 samples showed inhibition zones between 14 mm(20%) and 26 mm(20%) against *Klebsiella*, *Pseudomonas*, and *Bacillus* spp. [**Chart 3.2 / 3.3**] . Port Sudan sample showed moderate activity with inhibition zones between 16 mm(20%) and 20 mm(20%), varying by pathogen [**Chart 3.4**]. Nuri 1 (large) isolate exhibited the strongest antimicrobial activity, with an inhibition zone reaching 52 mm(20%) against *Staphylococcus aureus* [**Chart 3.6**].

Statistical analysis showed that the inhibition zones of soil isolates were significantly lower than control antibiotics ($p < 0.05$), except for Nuri 1 (large), which showed comparable or stronger activity in some cases [**Table 3.1**].

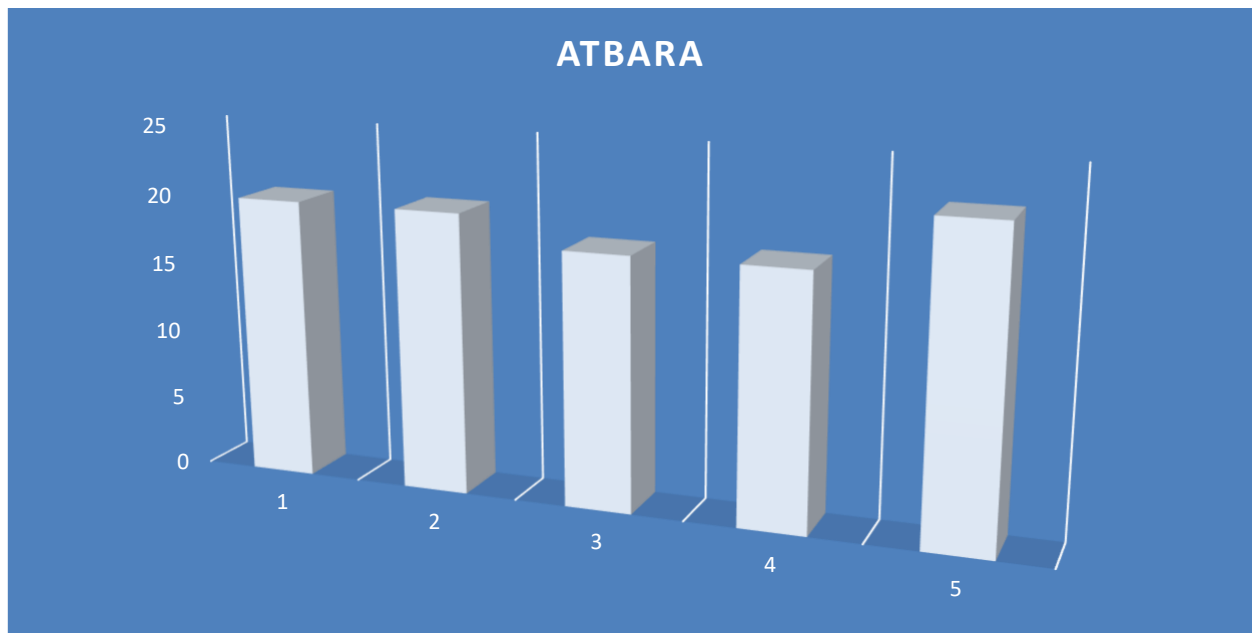


Chart 3.1 : The inhibition zone in Atbara value ranges between 18(40%) , 20(40%) and 22 (20%).

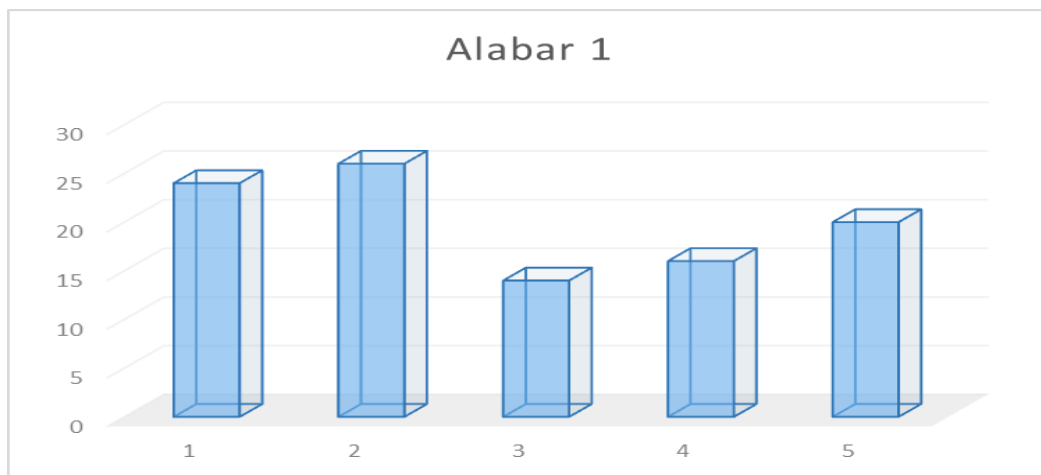


Chart 3.2 : The inhibition zone in Alabar 1 : value ranges between 14(20%) , 16(20%) , 20 (20%),24 (20%),26 (20%).

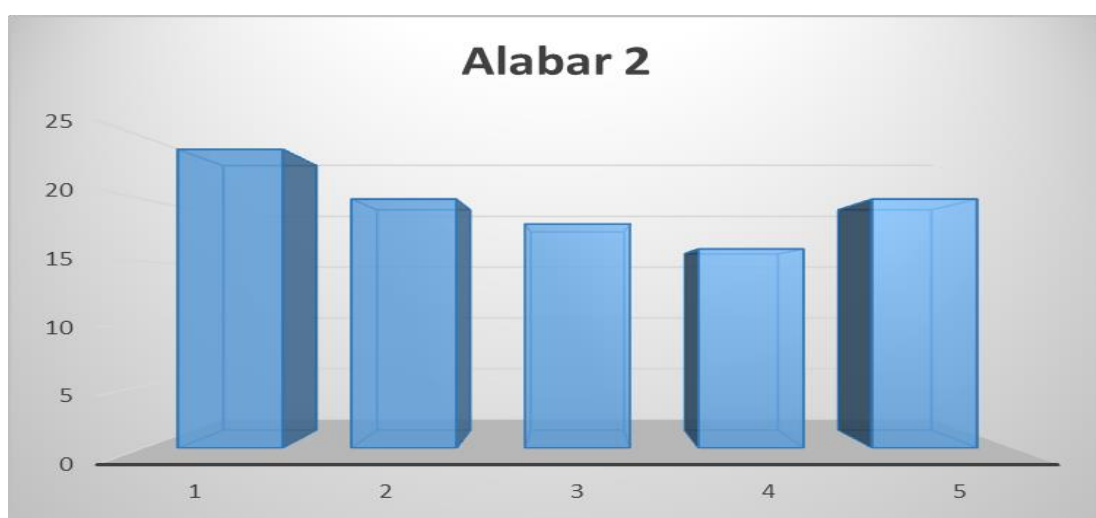


Chart 3.3 : The inhibition zone in Alabar 2 :value ranges between 16(20%) , 18(20%), 20(40%), 24(20%).

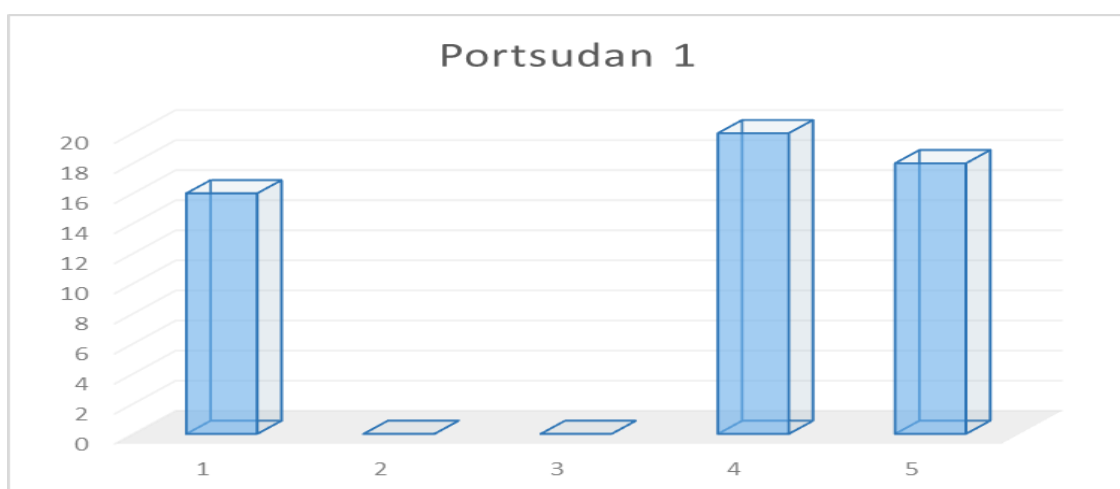


Chart 3.4 : The inhibition zone in Portsudan :value ranges between :0(40%) ,16(20%) , 18(20%), 20(20%).

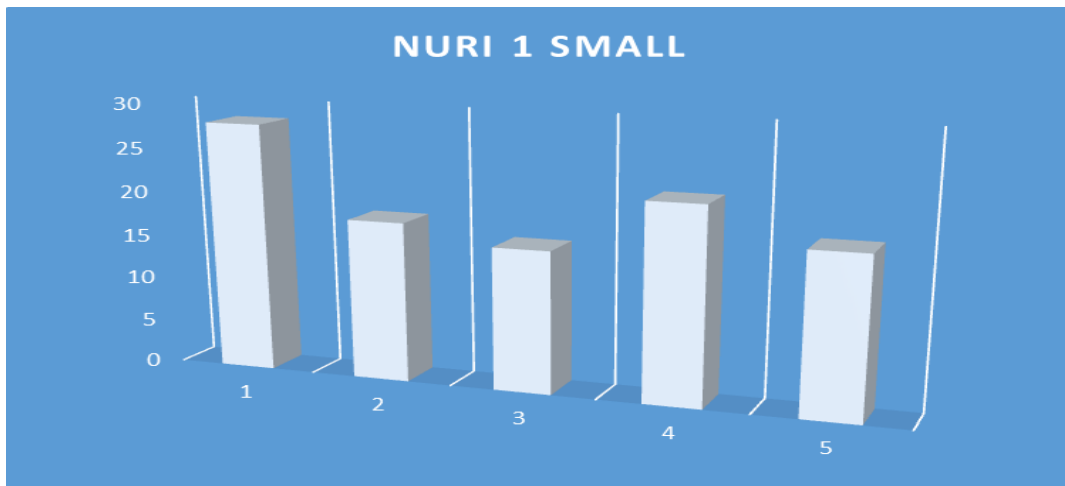


Chart 3.5 : The inhibition zone (Nuri1 small) :value ranges between :16(20%) ,18(40%) , 22 (20%),28 (20%).

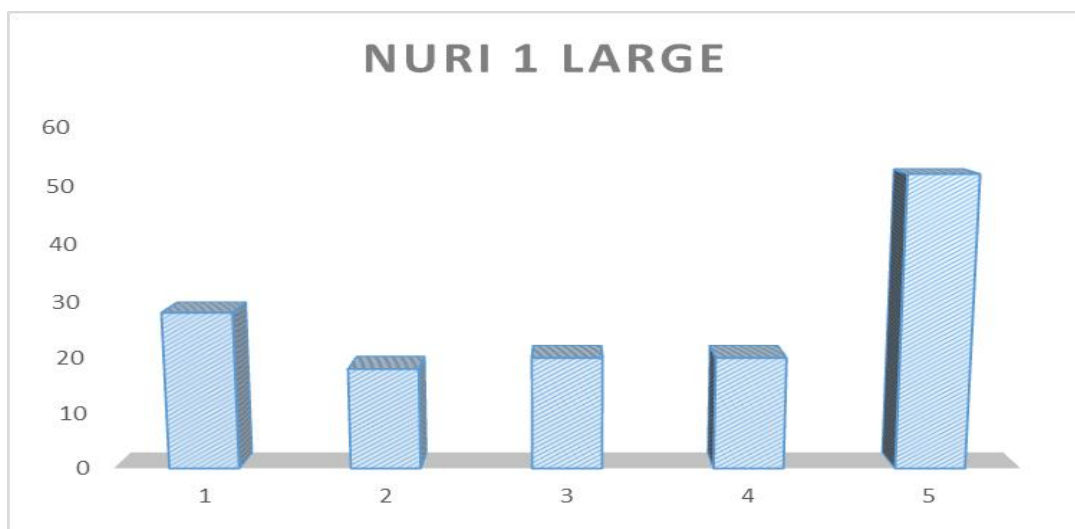


Chart 3.6 : The inhibition zone (Nuri1 large) :value ranges between : 18(20%) , 20(40%), 28(20%), 52(20%).

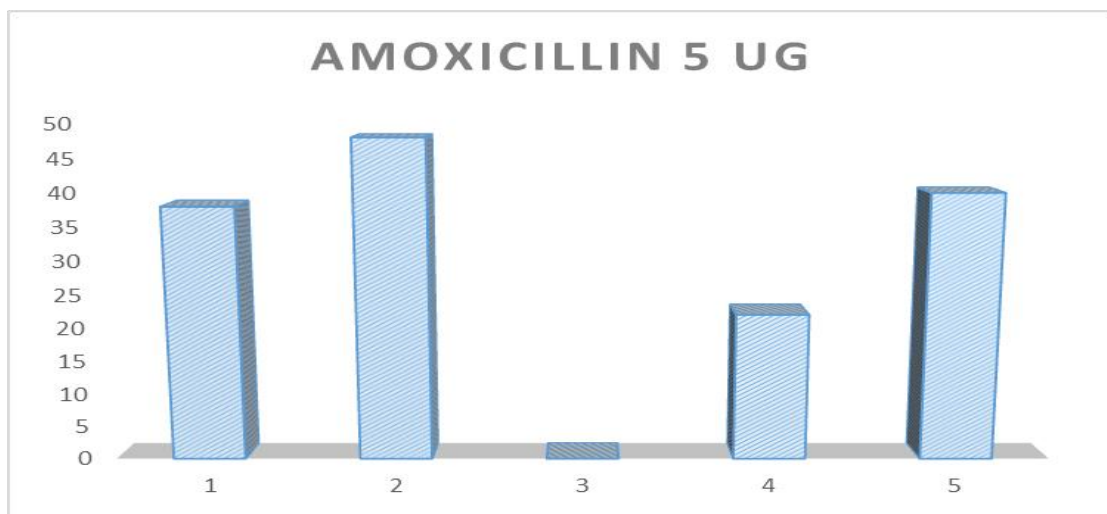


Chart 3.7 : The control antibiotic (Amoxicillin 5 ug): value ranges between : 0(20%) , 22(20%), 38(20%), 40(20%), 48(20%).

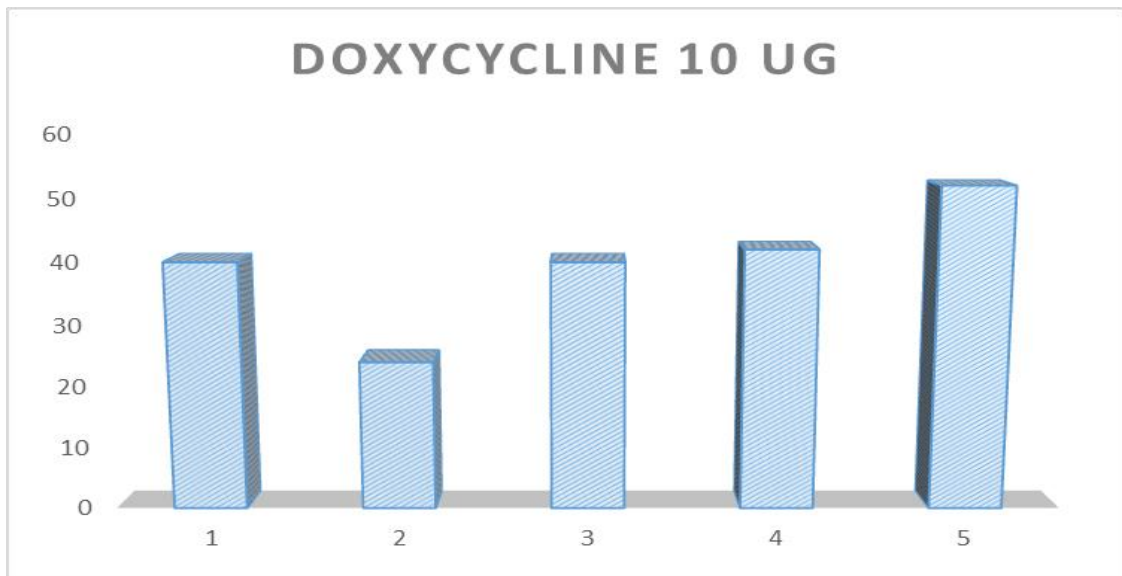


Chart 3.8 : The control antibiotic (Doxycycline 10 ug): value ranges between : 24(20%), 40(40%), 42(20%), 52(20%).

RESULTS

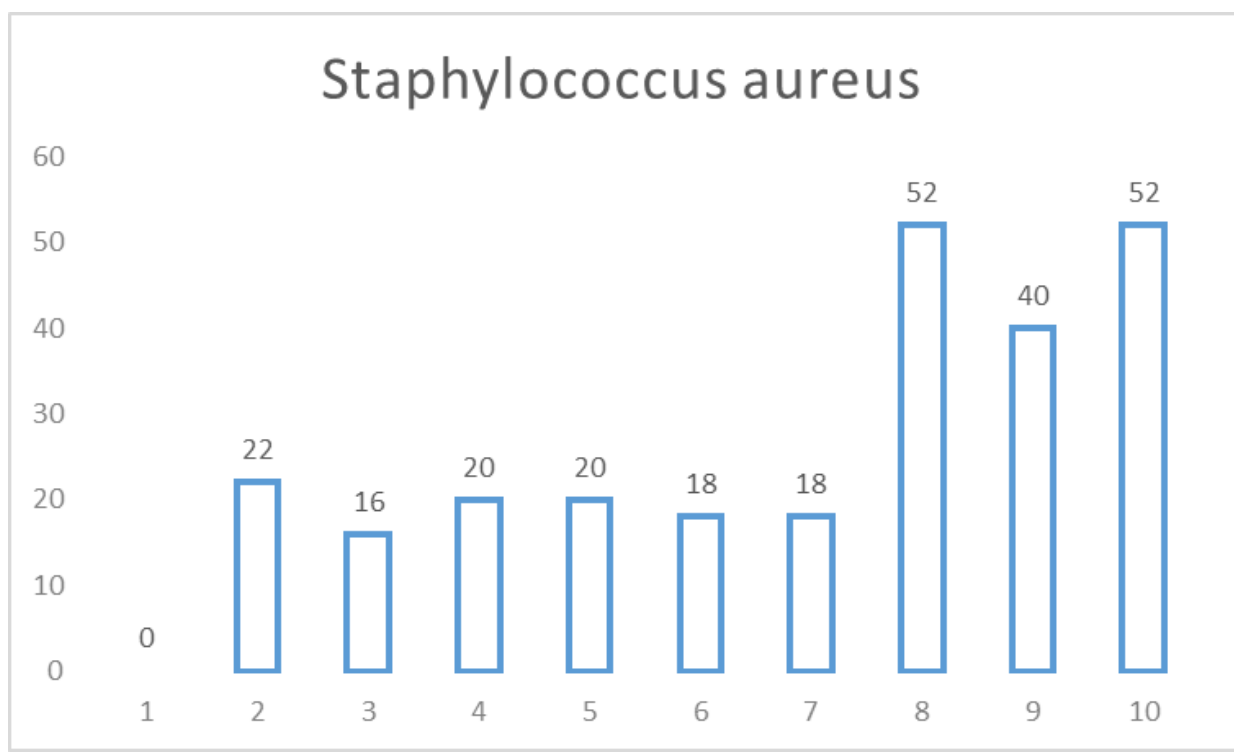


Chart 3.9 : Shown The bacterial strains used are clinical isolated (Staphylococcus aureus): value ranges between 0 and 52

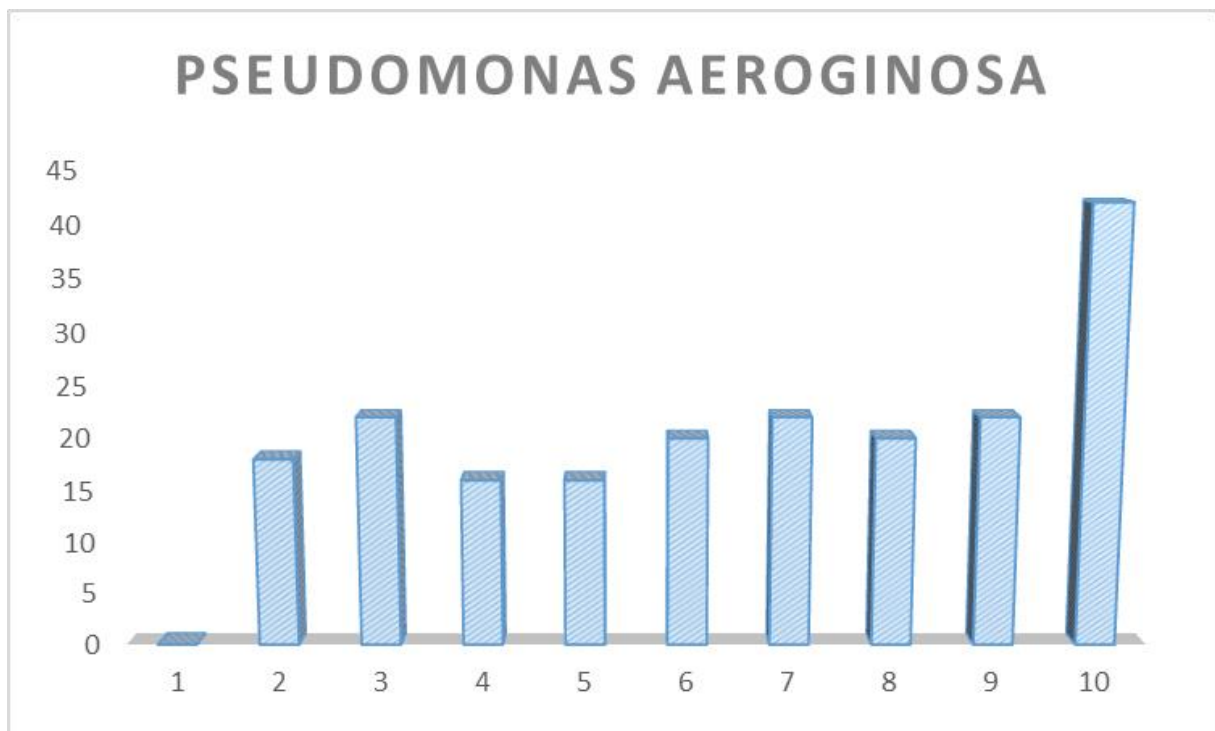


Chart 3.10 : The bacterial strains used are clinical isolated (*Pseudomonas aeruginosa*): value ranges between 0 and 22.

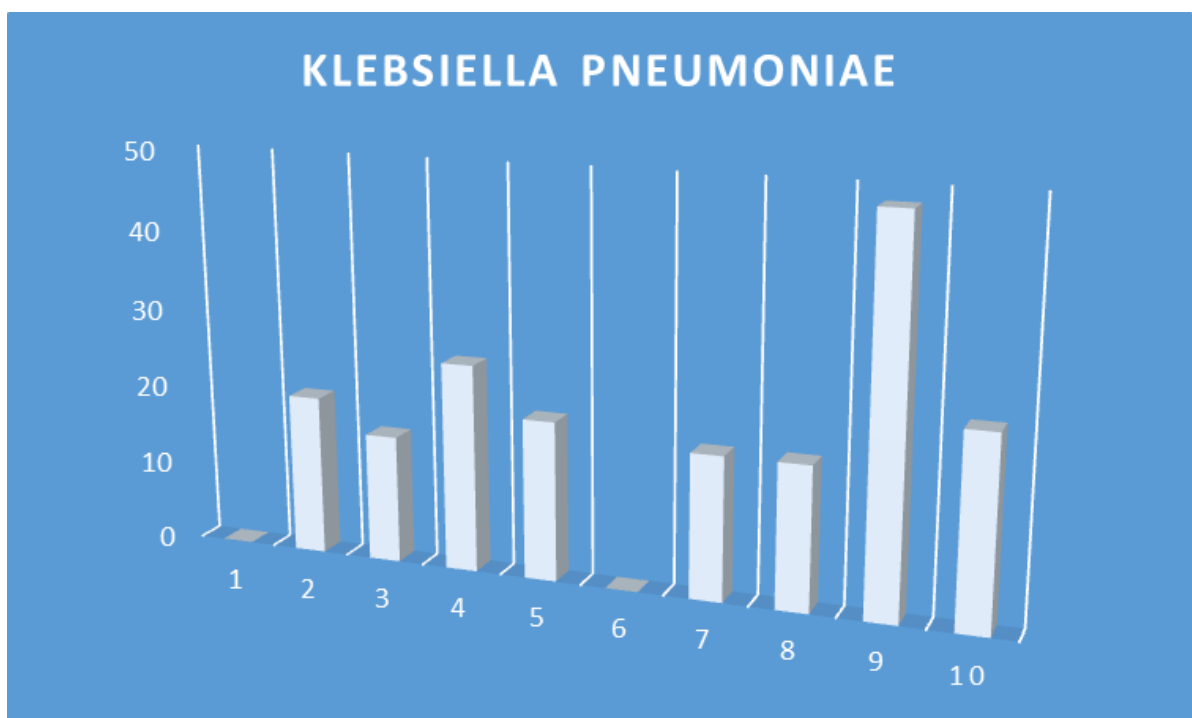


Chart 3.11 : The bacterial strains used are clinical isolated (*Klebsiella pneumoniae*) : value ranges between 0 and 26.

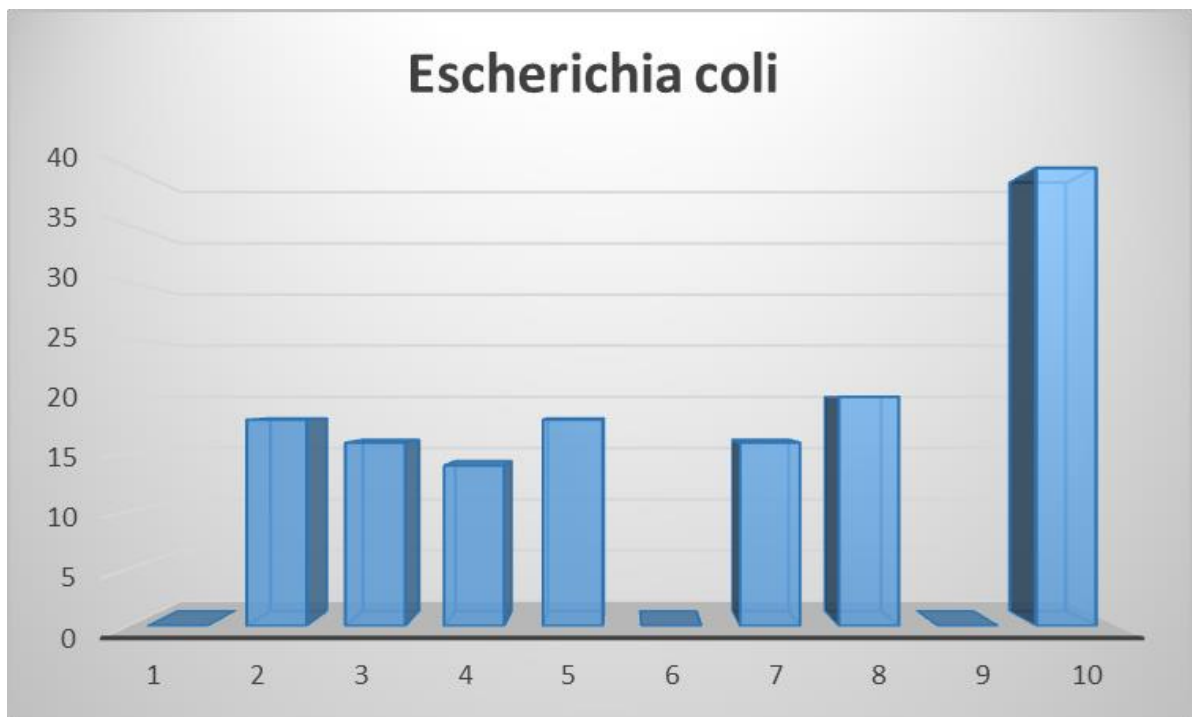


Chart 3.12 : The bacterial strains used are clinical isolated (*Escherichia coli*): value ranges between 0 and 20.

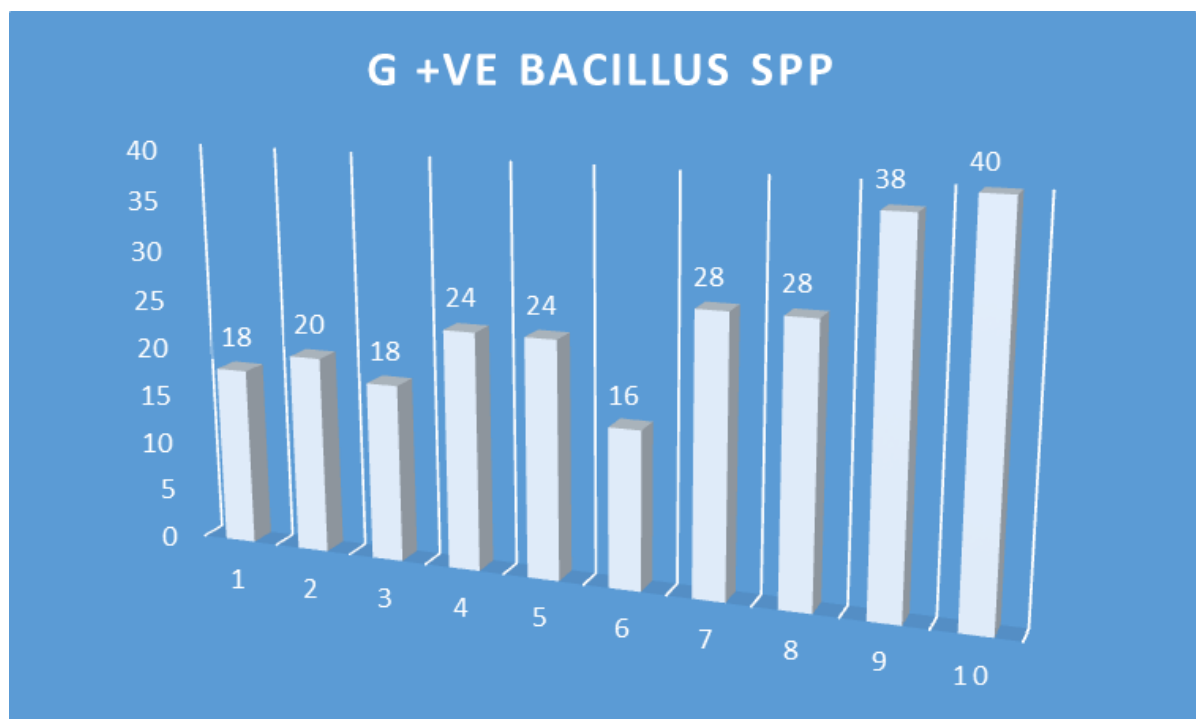


Chart 3.13 : The bacterial strains used are clinical isolated (Gram-positive *Bacillus* species) : value ranges between 16 and 28.

clinical Isolate	Independent(t-test)				p-value
	group	Descriptive statistic			
		minimum	mean	maximum	
G +ve bacillus spp	Case(n=8)	16	22	28	0.001
	Control(n=2)	38	39	40	
Klebsiella pneumoniae	Case	0	14.75	26	0.037
	Control	24	36	48	
Escherichia coli	Case	0	12.75	20	0.485
	Control	0	20	40	
Pseudomonas aeruginosa	Case	0	16.75	22	0.049
	Control	22	32	42	
Staphylococcus aureus	Case	0	20.75	52	0.049

Table 3.1 : Descriptive statistic of the bacterial strains used are clinical isolated (case-control).

Table (6) : Result:

1. The *G +ve bacillus spp* are significantly lower in case group compared to control group at significant 5%.
1. The *Klebsiella pneumoniae* are significantly lower in case group compared to control group at significant 5%.
2. The *Escherichia coli* are significantly no different between case groups compared to control group at significant 5%.
3. The *Pseudomonas aeruginosa* are significantly lower in case group compared to control group at significant 5%.
4. The *Staphylococcus aureus* are significantly lower in case group compared to control group at significant 5%.

Group	inhibition zone	clinical Isolate				
		G +ve	Klebsiella pneumon	E.coli	Pseudomonas aeroginosa	Staphy.aureus
Case	Sikka sita	18	0	0	0	0
	Atbara	20	20	18	18	22
	Shandi	18	16	16	22	16
	Alabar 1	24	26	14	16	20
	Alabar 2	24	20	18	16	20
	Port Sudan	16	0	0	20	18
	Nuri 1 small	28	18	16	22	18
	Nuri 1 large	28	18	20	20	52
Control	Amoxicillin 5 ug	38	48	0	22	40
	Doxycycline 10 ug	40	24	40	42	52
P-value		0. 267	0. 267	0. 281	0. 281	0. 267

Table 3.2 : The bacterial strains used are clinical isolated (case, control) with inhibition zone

Result:

(1)There is no affect from inhibition zone to the bacterial strains used are clinical isolated (case, control) **at significant 5%.**

Chapter four

Chapter Four:

4.1 Discussion

- This study investigated the antimicrobial potential of soil-derived microorganisms collected from multiple locations across *Sudan* . The findings demonstrate the significant presence of antibiotic-producing organisms, particularly *Bacillus* species and certain fungi, which exhibited varying degrees of inhibitory activity against clinically isolated bacterial pathogens.
- In our study the isolate from Nuri 1 (large) was the most potent, exhibiting an inhibition zone of 52 mm against *Staphylococcus aureus*, which is comparable to the zone produced by Amoxicillin (5 µg) in the control (40 mm). This aligns with the findings of (Elbendary et al., 2018), who reported several isolates from agriculture lands from *Egypt* showing lower activity (Average range : 5 – 10 mm) against *Staphylococcus aureus* strains, indicating that soil from less disturbed environments may harbor highly bioactive strains .
- In our study we used well diffusion method with glucose-enriched Mueller-Hinton agar that increased antimicrobial activity as seen in **Fig (3)** while (Elbendary et al., 2018) employed streak-plating technique on Mueller Hinton agar .
- Both of our studies used the Dutch streaking method for the detection of antimicrobial activity but the difference that in our study the target microorganisms that used are clinical isolated that showed high resistance rate among the commercial antibiotics while (Elbendary et al., 2018) used standard strains of each bacterial spp. .

4.2 Conclusion

- The study confirmed that soil remains a valuable reservoir for antibiotic-producing microorganisms. Isolates from locations such as Nuri, Alabar, Atbara, and Port Sudan demonstrated antimicrobial activity against multiple clinical pathogens, particularly Gram-positive bacteria like *Staphylococcus aureus* and *Bacillus* spp.

- Fungal isolates and *Bacillus* species were among the most effective antibiotic producers. While their activity was generally lower than commercial antibiotics, several isolates exhibited comparable or superior inhibition zones against selected bacteria.
- These findings validate the potential of environmental microbiota as a source of new antimicrobial agents, especially in the context of rising antibiotic resistance.

4.3 Recommendation

- Perform molecular identification (e.g., 16S rRNA sequencing) of the most potent isolates to precisely classify and compare them with known antibiotic-producing strains.
- Extract and chemically characterize the active compounds responsible for antimicrobial activity using chromatography and spectroscopy techniques.
- Test isolates against a wider range of pathogens, including multidrug-resistant strains, to evaluate their broad-spectrum efficacy.
- Optimize growth conditions (e.g., pH, temperature, nutrient composition) to maximize antibiotic production.
- Partner with research institutions and pharmaceutical companies for further development and clinical testing.
- Collect more soil samples from diverse ecosystems (e.g., deserts, forests, marine environments) to explore untapped microbial diversity.

4.4 Time Schedule :

Task / month	From 5 April to 27 June			
Proposal preparation	5April - 12 April			
Data collection		13April – 30 April		
Data analysis			1May - 5June	
Report writing and presenting for discussion				6June - 27June

4.5 Budget:

Item	Price
Nutrient broth 100g	450
Nutrient agar 100g	400
Sabroud dextrose agar 100g	520
Mulluer Hintone agar 100g	400
Yeast extract agar 500	2000
Plastic petri dishes 220 pcs	1100
Distilled water / Tubes container 20 pcs	50
Gauze / Cotton / Filter paper 11cm	200
Cylindrical measured tube 1L / Glass flask 500ml	340
KOH 50g	75
Gram stain set	200
Coverglass / slides	75
Pasture pipette 25 pcs	35
Cork borer	650
Falcon precipitating tubes 50 pcs	175
Automated pipette 10 – 100 uL / Yellow tips	810
Total	7480

4.6 Figures:

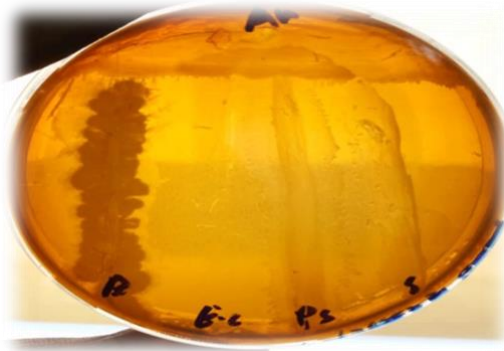


Fig 4.6.1 : Show the **Dutch Streak Method** used for detect antimicrobial produced from soil affect in clinical isolated bacterial strain.

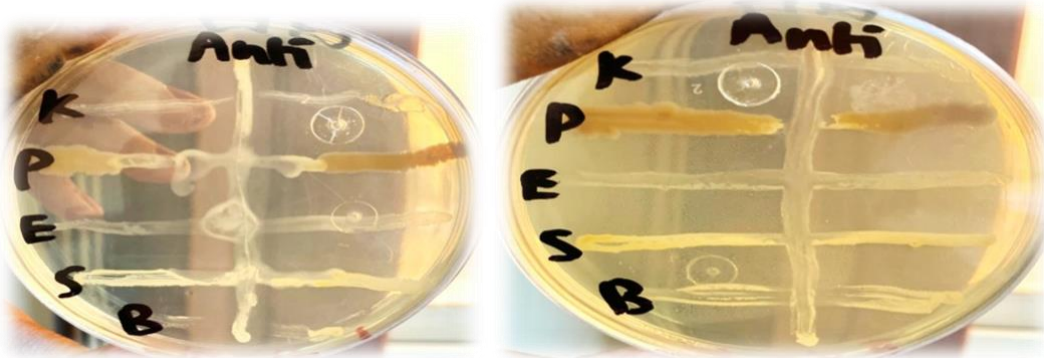


Fig 4.6.2 : show the **Modified "Dutch Method"** as confirmatory method to detect the antimicrobial effect.



Fig 4.6.3 : Show the **Antimicrobial Sensitivity Method** used to detect the effect of antimicrobial producer which isolated from the soil that has a synergism with the control antibiotic.

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Appendix



Cork borer device